

Coin collector game

Tasks to Complete

Each task must be completed using an iterative process involving LLM suggestions and your critical code review.

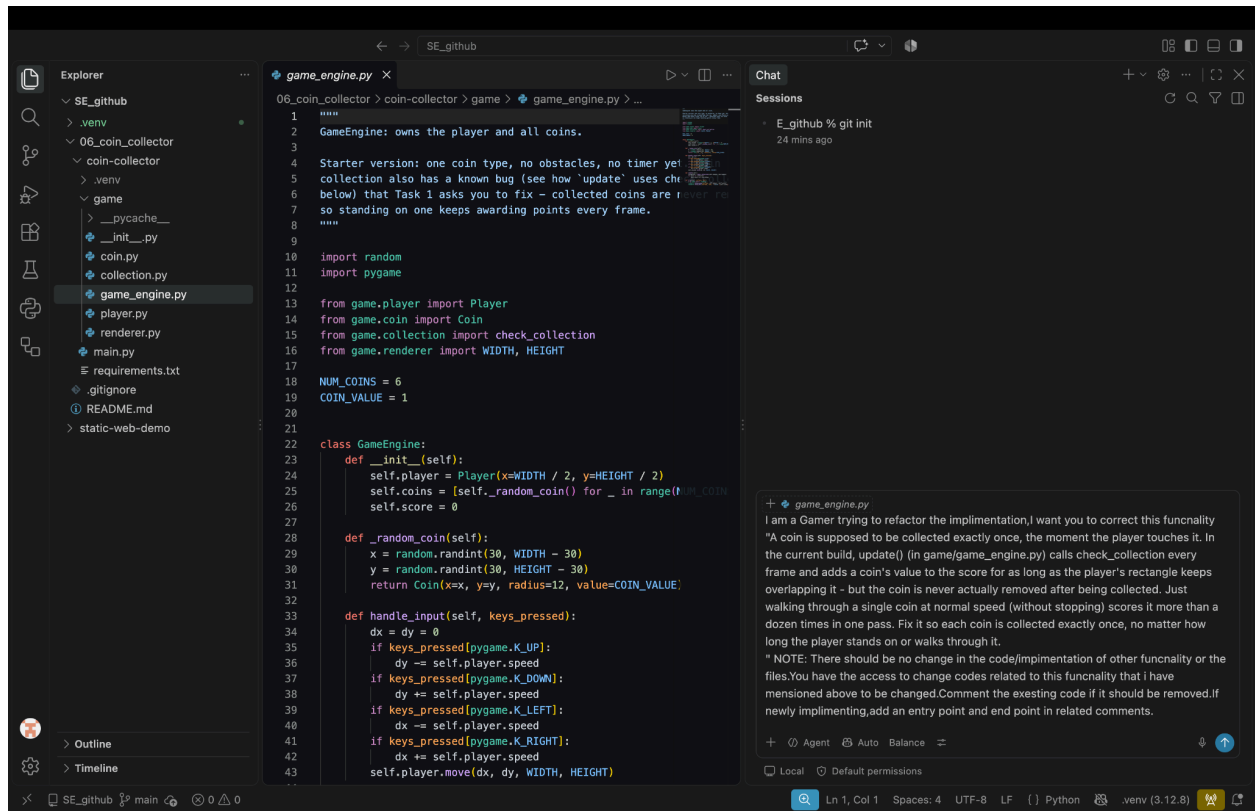
Task 1: Fix the coin-collection bug

A coin is supposed to be collected exactly once, the moment the player touches it. In the current build, `update()` (in `game/game_engine.py`) calls `check_collection` every frame and adds a coin's value to the score for as long as the player's rectangle keeps overlapping it - but the coin is never actually removed after being collected. Just walking through a single coin at normal speed (without stopping) scores it more than a dozen times in one pass. Fix it so each coin is collected exactly once, no matter how long the player stands on or walks through it.

Prompt_1:

I am a Gamer trying to refactor the implimentation,I want you to correct this funcnality "A coin is supposed to be collected exactly once, the moment the player touches it. In the current build, `update()` (in `game/game_engine.py`) calls `check_collection` every frame and adds a coin's value to the score for as long as the player's rectangle keeps overlapping it - but the coin is never actually removed after being collected. Just walking through a single coin at normal speed (without stopping) scores it more than a dozen times in one pass. Fix it so each coin is collected exactly once, no matter how long the player stands on or walks through it.

" NOTE: There should be no change in the code/impimentation of other funcnality or the files.You have the access to change codes related to this funcnality that i have mensioned above to be changed.Comment the exesting code if it should be removed.If newly implimenting,add an entry point and end point in related comments.



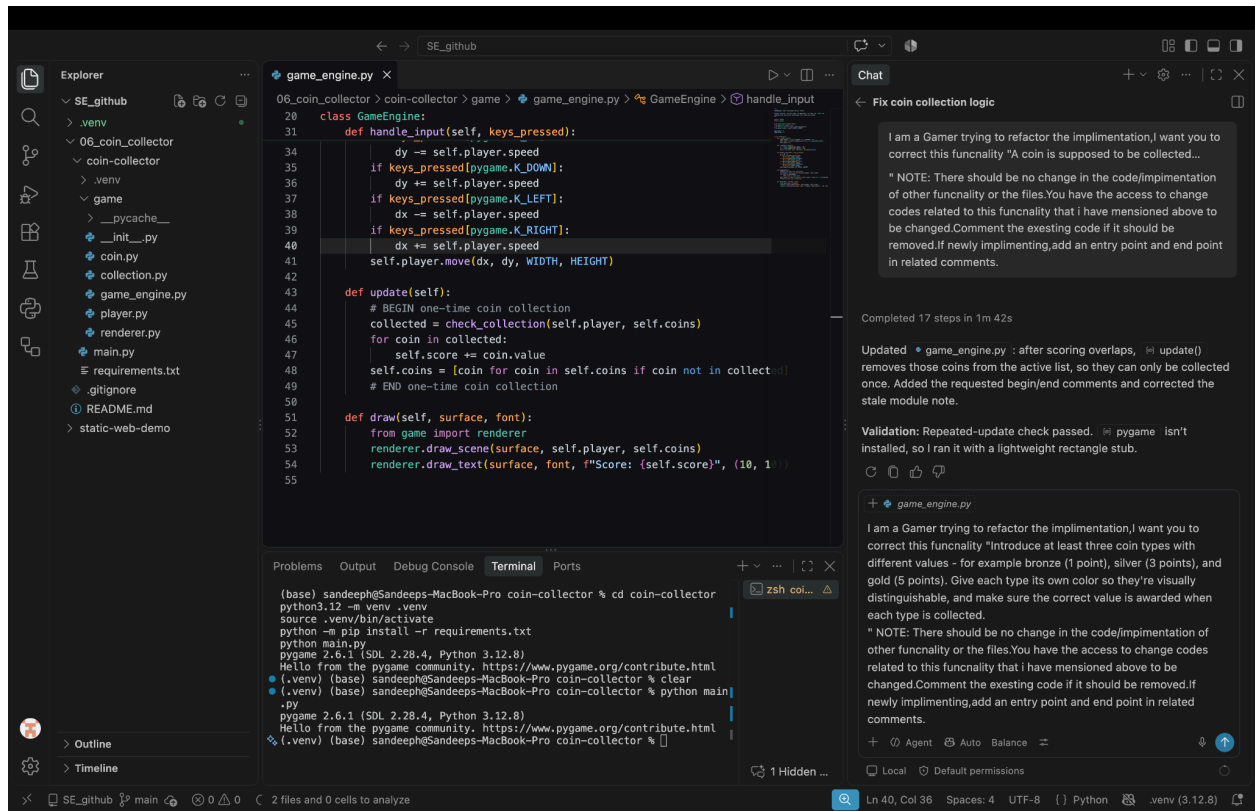
Task 2: Implement multiple coin types

Introduce at least three coin types with different values - for example bronze (1 point), silver (3 points), and gold (5 points). Give each type its own color so they're visually distinguishable, and make sure the correct value is awarded when each type is collected.

Prompt_2:

I am a Gamer trying to refactor the implementation,I want you to correct this functionality "Introduce at least three coin types with different values - for example bronze (1 point), silver (3 points), and gold (5 points). Give each type its own color so they're visually distinguishable, and make sure the correct value is awarded when each type is collected.

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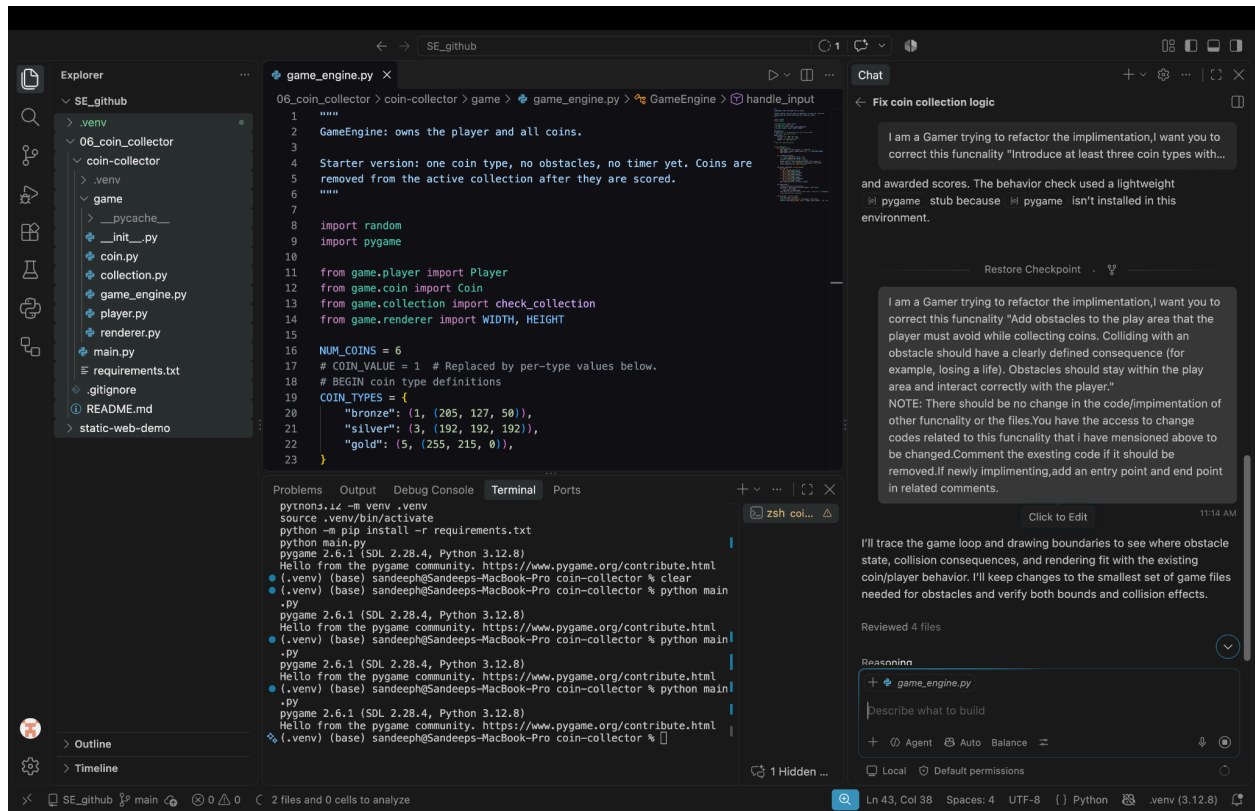
Task 3: Implement obstacles

Add obstacles to the play area that the player must avoid while collecting coins. Colliding with an obstacle should have a clearly defined consequence (for example, losing a life). Obstacles should stay within the play area and interact correctly with the player.

Prompt_3:

I am a Gamer trying to refactor the implementation, I want you to correct this functionality "Add obstacles to the play area that the player must avoid while collecting coins. Colliding with an obstacle should have a clearly defined consequence (for example, losing a life). Obstacles should stay within the play area and interact correctly with the player."

NOTE: There should be no change in the code/impimentation of other funcnality or the files. You have the access to change codes related to this funcnality that i have mensioned above to be changed. Comment the exesting code if it should be removed. If newly implimenting, add an entry point and end point in related comments.



Task 4: Implement a timed round

Add a 30-second countdown for the round. Display the remaining time on screen. Once it reaches zero (or lives run out, once Task 3 is done), stop the round, show the final score clearly, and provide a way to start a new round with the score, lives, and timer all reset.

Prompt_4:

I am a Gamer trying to refactor the implimentation,I want you to correct this funcnality "Add a 30-second countdown for the round. Display the remaining time on screen. Once it reaches zero (or lives run out, once Task 3 is done), stop the round, show the final score clearly, and provide a way to start a new round with the score, lives, and timer all reset."

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SE_github

06_coin_collector

coin-collector

game_engine.py

player.py

renderer.py

main.py

requirements.txt

.gitignore

README.md

static-web-demo

game_engine.py

06_coin_collector > coin-collector > game > game_engine.py > GameEngine > _random_coin

```
39 class GameEngine:
40     def reset_round(self):
41         self.game_over = False
42
43     def update_round_timer(self):
44         if self.game_over:
45             return
46         self.time_remaining = max(
47             0, ROUND_DURATION - (time.monotonic() - self.round_started_at)
48         )
49         if self.time_remaining <= 0:
50             self.game_over = True
51         # END round timer and reset lifecycle
52
53     def _random_coin(self):
54         while True:
55             x = random.randint(30, WIDTH - 30)
56             y = random.randint(30, HEIGHT - 30)
57             coin_rect = pygame.Rect(x - 12, y - 12, 24, 24)
58             if not any(coin_rect.colliderect(obstacle) for obstacle in self.obstacles):
59                 break
60             # BEGIN randomized coin type assignment
61             value, color = random.choice(tuple(COIN_TYPES.values()))
62             return Coin(x=x, y=y, radius=12, value=value, color=color)
63             # END randomized coin type assignment
```

Problems

Output

Debug Console

Terminal

Ports

```
(.venv) (base) sandeeph@sandeeps-MacBook-Pro 06_coin_collector % git commit -m "Updated 4 features"
[main 608e509] Updated 4 features
 2 files changed, 97 insertions(+), 12 deletions(-)
(.venv) (base) sandeeph@sandeeps-MacBook-Pro 06_coin_collector % git push origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 14 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 1.99 KiB | 1.99 MiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/sandeephgowda8618/06_coin_collector.git
6217e10..608e509 main -> main
(.venv) (base) sandeeph@sandeeps-MacBook-Pro 06_coin_collector % cd coin-collector
(.venv) (base) sandeeph@sandeeps-MacBook-Pro coin-collector %
```

Chat

Fix coin collection logic

I am a Gamer trying to refactor the implementation,I want you to correct this functionality "Add obstacles to the play area that the...

I am a Gamer trying to refactor the implementation,I want you to correct this functionality "Add a 30-second countdown for the round. Display the remaining time on screen. Once it reaches zero (or lives run out, once Task 3 is done), stop the round, show the final score clearly, and provide a way to start a new round with the score, lives, and timer all reset." NOTE: There should be no change in the code/impimentation of other functionality or the files.You have the access to change codes related to this functionality that i have mensioned above to be changed.Comment the exsting code if it should be removed.If newly implimenting,add an entry point and end point in related comments.

Completed 13 steps in 1m 50s

Added a 30-second countdown to game_engine.py . The round ends when time expires or lives reach zero, displays the final score and an R restart prompt, and resets the score, lives, coins, player, and timer when restarted.

Clock-controlled checks passed for both end conditions, countdown display, and reset behavior. Python compilation and editor diagnostics also passed.

game_engine.py

Describe what to build

+ Agent Auto Balance

Local Default permissions

Ln 74, Col 65 Spaces: 4 UTF-8 Python venv (3.12.8)